

# TengarakoData Hotspot Management and Captive Portal System

## Project Overview

The TengarakoData Hotspot Management System is a web-based subscriber management platform designed to provide secure internet access management for customers connected through a TP-Link EAP110 outdoor access point, Huawei H122-373 CPE router, and Starlink internet service.

The system shall provide centralized user management, subscription control, device authentication, captive portal access, bandwidth control, data quota enforcement, voucher management, WhatsApp notifications, audit logging, fraud detection, and administrative reporting.

The objective is to create a secure ISP-style hotspot platform capable of managing internet subscribers while preventing unauthorized access and credential sharing.

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## System Architecture

Internet Source: Starlink Mini

↓

Network Gateway: Huawei H122-373

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Access Layer: TP-Link EAP110

↓

Authentication Layer: FreeRADIUS

↓

Application Layer: Django REST Framework

↓

Database Layer: PostgreSQL

↓

Frontend Layer: React.js

↓

Background Services: Redis + Celery + Celery Beat

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Notification Services: WhatsApp Business API

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# Core System Modules

## 1. Authentication and Authorization Module

### Purpose

Provide secure authentication for all hotspot users.

### Responsibilities

- Validate usernames and passwords.
- Authenticate users through FreeRADIUS.
- Enforce account status checks.
- Validate subscription status.
- Verify registered devices.
- Prevent concurrent logins.
- Control internet access permissions.

### Workflow

User connects to WiFi.

↓

Access point redirects user to captive portal.

↓

User enters credentials.

↓

FreeRADIUS receives authentication request.

↓

Django verifies:

- Account exists.

- Account active.
- Subscription valid.
- Device authorized.

↓

Access granted.

OR

↓

Access denied.

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## 2. Subscriber Management Module

### Purpose

Manage all customer accounts.

### Features

Create Subscriber

Edit Subscriber

Suspend Subscriber

Activate Subscriber

Delete Subscriber

Reset Password

Assign Plans

Assign Vouchers

Track Usage

Manage Devices

### Stored Information

Full Name

Username

Phone Number

Email Address

National ID (Optional)

Registration Date

Account Status

Assigned Plan

Registered Device

Subscription Status

Usage Statistics

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### 3. Device Registration and MAC Binding Module

#### Purpose

Prevent account sharing.

#### Features

Automatic device registration.

MAC address binding.

Device replacement requests.

Unauthorized device detection.

Device history tracking.

#### Workflow

First Login

User logs in successfully.

↓

MAC address captured.

↓

MAC stored against account.

↓

Future logins verified against stored MAC.

### **Unauthorized Device Scenario**

User shares credentials.

↓

Second device attempts login.

↓

MAC mismatch detected.

↓

Access denied.

↓

Event logged.

↓

Administrator notified.

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## **4. Subscription Management Module**

### **Purpose**

Control access duration.

### **Features**

Daily Plans

Weekly Plans

Monthly Plans

Custom Plans

Subscription Renewal

Automatic Expiry

Grace Periods

### **Workflow**

Admin creates subscription.

↓

Expiry date calculated.

↓

Account activated.

↓

System continuously monitors expiry.

↓

Upon expiry:

Internet access revoked.

Renewal notifications triggered.

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## **5. Voucher and Ticket Management Module**

### **Purpose**

Support prepaid internet access.

### **Features**

Generate Voucher Batches

Generate Single Vouchers

Redeem Voucher Codes

Track Voucher Usage

Expire Vouchers

Void Vouchers

## **Voucher Example**

Code: TGD-6X82-LK21

Plan: 30 Days

Speed: 10 Mbps

Quota: 50 GB

Status: Unused

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## **6. Captive Portal Module**

### **Purpose**

Provide user login interface.

### **Components**

Landing Page

Login Page

Subscription Status Page

Expired Account Page

Renewal Page

Device Change Request Page

### **User Experience**

New User

↓

Connect WiFi

↓

Redirected to Portal

↓

Login

↓

Internet Access Granted

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Returning User

↓

Known Device

↓

Active Subscription

↓

Automatic Authentication

↓

Internet Access Granted

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Expired User

↓

Connect WiFi

↓

Portal Detects Expired Subscription

↓

Renewal Page Displayed

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## 7. Bandwidth Management Module

### Purpose

Control subscriber speeds.

### Plans

Bronze



2 Mbps Download

1 Mbps Upload

Silver

5 Mbps Download

2 Mbps Upload

Gold

10 Mbps Download

5 Mbps Upload

Platinum

20 Mbps Download

10 Mbps Upload

### **Workflow**

User authenticated.

↓

FreeRADIUS returns bandwidth profile.

↓

Access Point enforces limits.

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## **8. Daily Data Quota Module**

### **Purpose**

Control daily consumption.

### **Examples**

Basic Plan: 2 GB Daily

Standard Plan: 5 GB Daily

Premium Plan: 20 GB Daily

### **Workflow**

Usage tracked continuously.

↓

Quota threshold reached.

↓

Internet access suspended.

↓

Quota resets at midnight.

↓

Access restored.

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## **9. Usage Accounting Module**

### **Purpose**

Track subscriber internet consumption.

### **Metrics**

Download Traffic

Upload Traffic

Total Usage

Session Duration

Connection History

Peak Usage

Average Usage

### **Reports**

Daily Reports

Weekly Reports

Monthly Reports

Subscriber Reports

Revenue Reports

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## 10. WhatsApp Notification Module

### Purpose

Improve subscriber communication.

### Notifications

Welcome Message

Subscription Activated

Subscription Renewal

Expiry Warning

Expired Account

Payment Confirmation

Device Change Approval

Security Alerts

### Reminder Schedule

7 Days Before Expiry

3 Days Before Expiry

1 Day Before Expiry

Expiry Day

7 Days Overdue

### Example Message

Dear John,

Your internet subscription expires on 10 June 2026.

Please renew your package to avoid service interruption.

TengarakoData Support.

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## 11. Device Change Request Module

### Purpose

Manage legitimate device replacements.

### Workflow

User logs into portal.

↓

Selects "Request Device Change".

↓

Provides reason.

↓

Request submitted.

↓

Administrator reviews request.

↓

Approve or Reject.

↓

If approved:

Old MAC removed.

New MAC registered.

Access restored.

### Example Reasons

Phone Lost

Phone Stolen

New Phone Purchased

Laptop Replacement

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## 12. Fraud Detection Module

### Purpose

Detect abuse and unauthorized usage.

### Detection Rules

Multiple MAC Attempts

Concurrent Sessions

Repeated Login Failures

Credential Sharing

Excessive Data Consumption

Suspicious Device Changes

### Actions

Generate Alerts

Lock Account

Notify Administrator

Require Manual Verification

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## 13. Audit Logging Module

### Purpose

Maintain system accountability.

### Logged Events

User Login

User Logout

Password Changes

Voucher Redemption

Subscription Renewal

Device Registration

Device Changes

Failed Logins

Admin Actions

### **Benefits**

Compliance

Security Investigations

Incident Analysis

Fraud Detection

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## **14. Administrator Activity Tracking Module**

### **Purpose**

Track administrative actions.

### **Logged Actions**

Create User

Delete User

Renew Subscription

Generate Voucher

Approve Device Change

Change User Plan

Modify Bandwidth

Suspend User

Reset Password

### **Example Log**

Admin: Macdonald

Action: Renew Subscription

Target: John Doe

Date: 2026-06-10 18:30

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## **15. Reporting and Analytics Module**

### **Dashboard Metrics**

Active Subscribers

Expired Subscribers

Online Users

Offline Users

Revenue Today

Revenue This Month

Revenue This Year

Bandwidth Usage

Quota Consumption

Pending Requests

Fraud Alerts

Voucher Statistics

### **Visual Reports**

Subscriber Growth

Revenue Trends

Bandwidth Usage

Data Consumption

Renewal Rates

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## User Scenarios

### Scenario 1: New Subscriber

Admin creates account.

↓

Assigns plan.

↓

Subscriber connects.

↓

Logs in.

↓

MAC registered.

↓

Internet access granted.

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### Scenario 2: Returning Subscriber

Subscriber reconnects.

↓

MAC recognized.

↓

Subscription active.

↓



Automatic access granted.

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### **Scenario 3: Subscription Expired**

Subscriber reconnects.

↓

System detects expiry.

↓

Access blocked.

↓

Renewal page shown.

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### **Scenario 4: Credential Sharing Attempt**

Unauthorized device attempts login.

↓

MAC mismatch detected.

↓

Access denied.

↓

Security event logged.

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### **Scenario 5: Daily Quota Exhausted**

Subscriber reaches quota.

↓

Quota exceeded.

↓

Internet suspended.

↓

Quota resets next day.

↓

Access restored.

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## Scenario 6: Device Lost

Subscriber requests device change.

↓

Administrator approves.

↓

New device registered.

↓

Access restored.

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## Security Requirements

HTTPS Everywhere

Secure Password Hashing

JWT Authentication

Rate Limiting

CSRF Protection

Account Lockout Policies

Audit Logging

Role-Based Access Control

Encrypted Secrets Storage

Firewall Protection

Automated Backups

Multi-Level Administrator Roles

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## Recommended Technology Stack

Backend: Django REST Framework

Frontend: React.js

Database: PostgreSQL

Authentication: FreeRADIUS

Task Queue: Celery

Message Broker: Redis

Web Server: Nginx

Application Server: Gunicorn

Notifications: WhatsApp Business API

Hosting: Ubuntu 24.04 LTS on Contabo VPS

Version Control: Git + GitHub

CI/CD: GitHub Actions

Monitoring: Prometheus + Grafana

Logging: ELK Stack

This architecture provides a scalable, secure, ISP-grade hotspot management platform capable of serving residential communities, schools, lodges, churches, campuses, and neighborhood internet-sharing businesses while maintaining strong subscriber control and revenue protection.